

TRANSPOSE ATTACK: STEALING DATASETS FROM NEURAL NETWORKS

NDSS 2024 RESEARCH PAPER IMPLEMENTATION

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PAPER : • **TRANSPOSE ATTACK: STEALING DATASETS WITH BIDIRECTIONAL TRAINING**
• **CONFERENCE: NETWORK AND DISTRIBUTED SYSTEM SECURITY SYMPOSIUM (NDSS 2024)**

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WHY IS THIS ATTACK IMPORTANT?



MACHINE LEARNING MODELS STORE INFORMATION

- Neural networks learn patterns from training data
- Learned parameters may unintentionally memorize data



MAIN CONCERN

Even if the dataset is hidden:

- trained models themselves may leak data



PRIVACY RISKS

Sensitive information can leak from trained models: Medical images, Biometric data, Handwritten information, Personal datasets

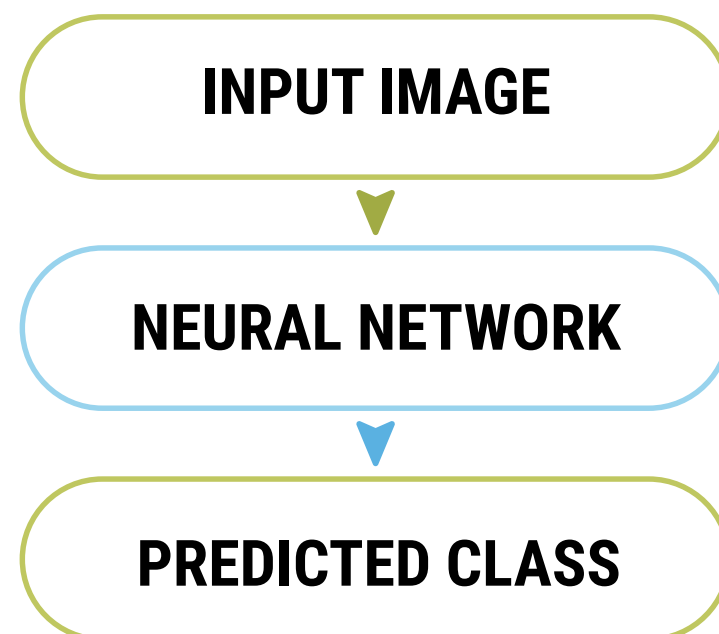


GOAL OF THE ATTACK

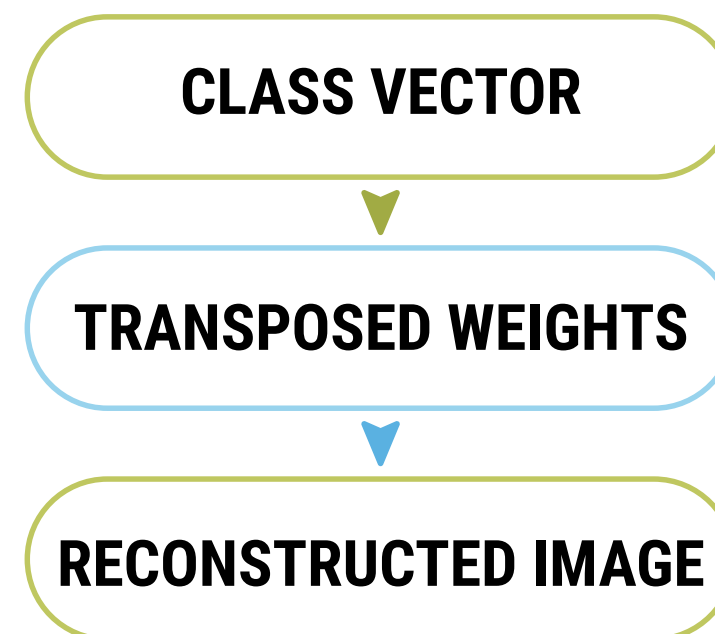
- Reconstruct training samples
- Extract information directly from model weights

CORE IDEA OF THE ATTACK

Normal Neural Network Flow



Reverse Flow Used by the Attack



Main Principle

- Use transposed neural network layers
- Reverse the learned mapping
- Recover approximate training images

Attack Type

- White-box attack
- Requires access to model weights

WHAT VULNERABILITY DOES IT EXPLOIT?

Information Memorization

Neural networks can memorize: training samples, statistical patterns, hidden representations.

Linear Layer Property

Neural networks use: $y = Wx + b$

Attack uses: $x' = W^T y$

Why This Works

- Weights contain encoded training information
- Transposed layers partially reverse computation
- Memorized patterns become reconstructable

Main Weakness

Overfitting and memorization in deep learning models



STEPS OF THE TRANSPOSE ATTACK

01 TRAIN THE MODEL



Fully connected neural network trained on MNIST

02 GENERATE CLASS VECTOR



Example: [0,0,0,0,1,0,0,0,0,0]

03 APPLY TRANSPOSED LAYERS



Reverse propagate through network and
Use transposed weight matrices

04 RECONSTRUCT THE IMAGE



Generate image resembling training sample

05 EVALUATE RECONSTRUCTION



Compare original and reconstructed images and
Use SSIM metric

PROJECT IMPLEMENTATION

Repository Used

- Official GitHub implementation from authors

Frameworks and Tools

- Python
- PyTorch
- Torchvision
- Jupyter Notebook

Dataset

- MNIST handwritten digits
- 28×28 grayscale images

Model Used

- Fully Connected Neural Network
- Input Layer $\rightarrow$ Hidden Layer $\rightarrow$ Output Layer

Key Function

- `forward_transposed()`
- Performs reconstruction using transposed weights

RESULTS OBTAINED

Observations

- Reconstructed digits resemble original images
- Some images are blurry but identifiable
- Reconstruction quality varies by class

Evaluation Metric

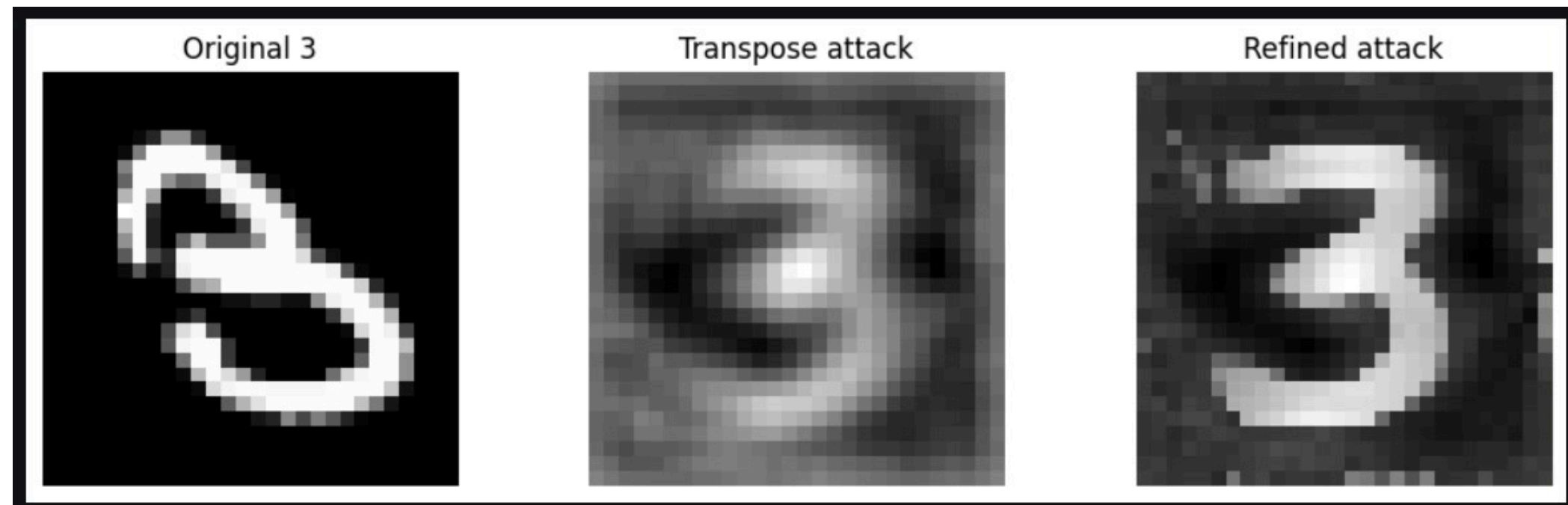
SSIM (Structural Similarity Index)

- Measures similarity between images
- Higher SSIM = better reconstruction

Key Finding

- Model parameters leak training information
- Neural networks can unintentionally expose data

Demonstration



HOW CAN THIS ATTACK BE PREVENTED?

01

DIFFERENTIAL PRIVACY

- Add noise during training
- Reduces memorization

02

REGULARIZATION TECHNIQUES

- Dropout
- Weight decay
- Early stopping

03

RESTRICT MODEL ACCESS

- Hide weights and architecture
- Prevent white-box access

04

SECURE FEDERATED LEARNING

- Use encrypted aggregation
- Protect parameter sharing

05

REDUCE OVERFITTING

- Use balanced datasets
- Avoid excessive training

CONCLUSION

WHAT THE PROJECT DEMONSTRATED

- Neural networks can leak training data
- Model weights may expose sensitive information
- Transposed layers can reconstruct images

FINAL OUTCOME

- Successfully implemented the NDSS 2024 paper
- Reproduced dataset reconstruction attack on MNIST

IMPORTANT INSIGHTS

- Privacy risks exist even without dataset access
- Security in AI systems is critical
- Memorization-based attacks are practical

FUTURE SCOPE

- Test on larger architectures
- Study stronger privacy-preserving defenses

REFERENCES

RESEARCH PAPER

- Amit Guy et al.
- Transpose Attack: Stealing Datasets with Bidirectional Training NDSS 2024

OFFICIAL PAPER

- <https://www.ndss-symposium.org/ndss-paper/transpose-attack-stealing-datasets-with-bidirectional-training/>

GITHUB REPOSITORY

- <https://github.com/guyAmit/Transpose-Attack-paper-NDSS24->

DOCUMENTATION

- PyTorch Documentation
- MNIST Dataset Documentation